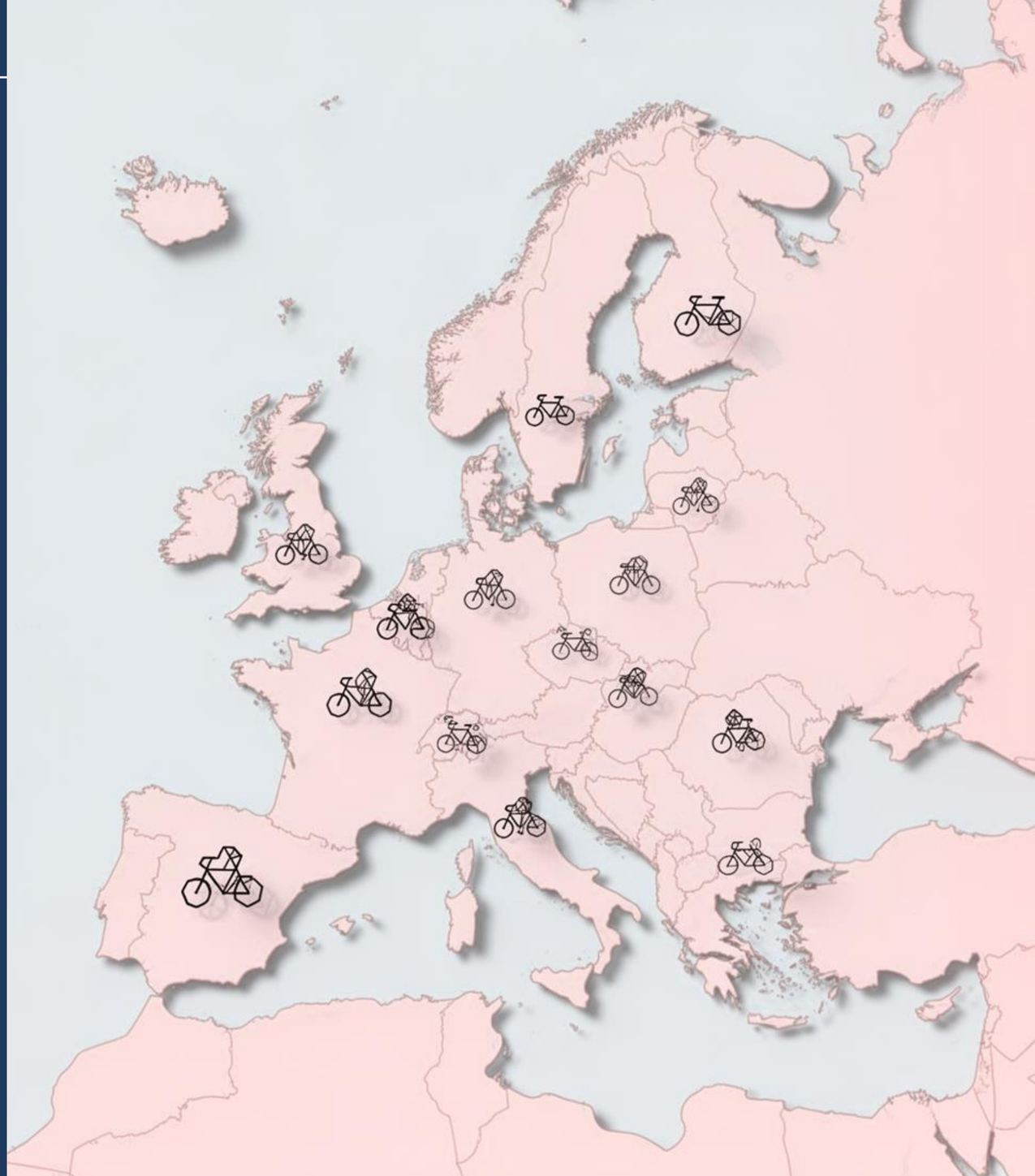
Bike

Sales Analysis

Prepared by Merna Mounir; March 24, 2026



Executive Summary

This project presents an exploratory data analysis of a bike sales dataset in Europe.
The objective is to identify key trends in revenue,profit,customer segments,and product performance.
The analysis highlights seasonal patterns, high-performing categories,and geographical differences.
Insights are derived to support data-driven business decisions and strategic planning.

Data Overview

Data Source:

Bike sales dataset

Data Quality:

Missing values were checked
Duplicate records were identified and removed
Data types were consistent
Dataset cleaned and prepared for analysis

Data Size

Number of rows: 113036
Number of columns: 18

Key Performance Indicators

Most Common Gender:
The gender that appears most frequently among customers.

Total Profit:
The total profit earned from all sales transactions.

Revenue by Country:
Total revenue generated from each country.

Top Product Category:
The product category with the highest number of purchases.

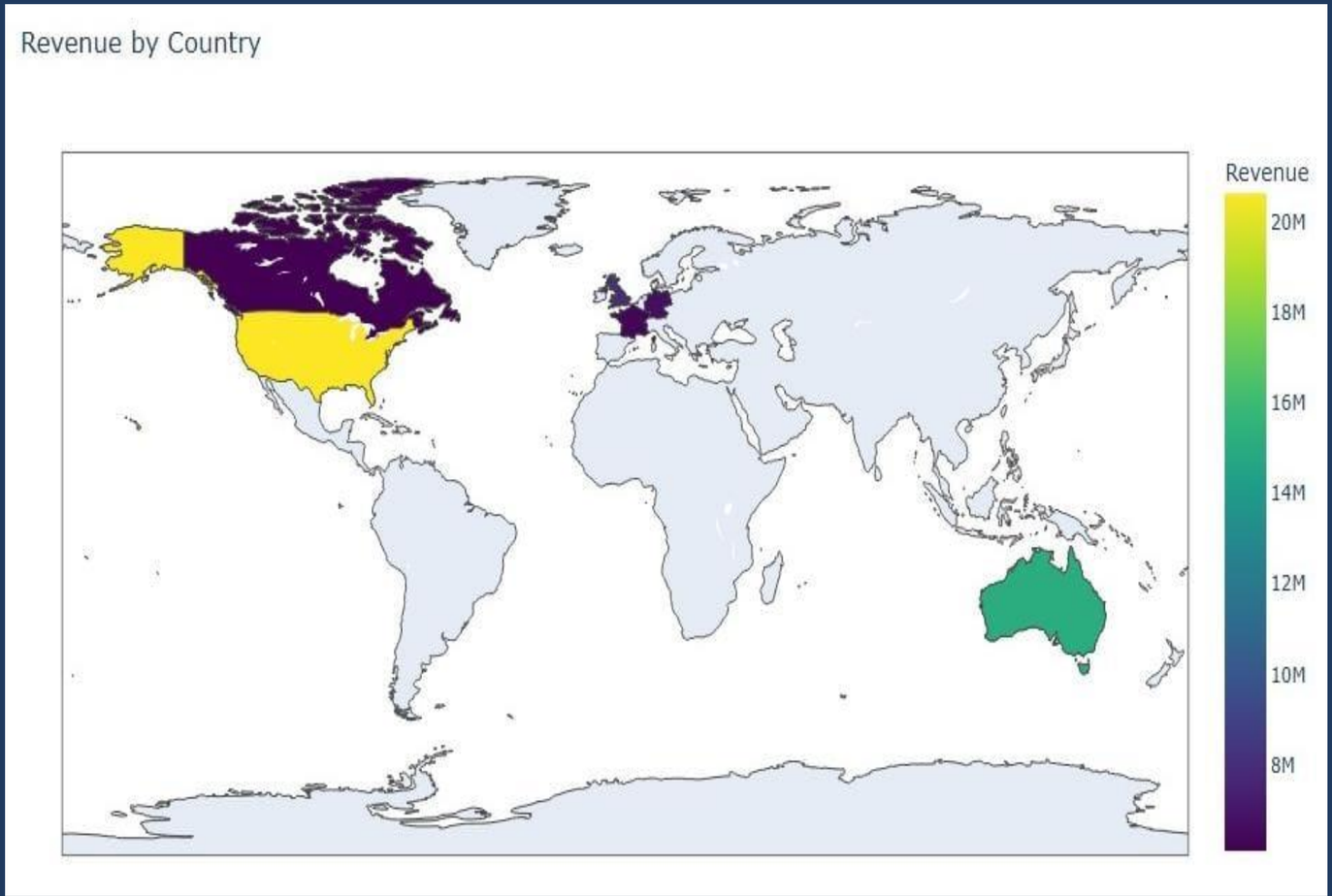
Revenue by Time (Year/Month):
Sales performance over time based on year and month.

Customer Gender Distribution:
Distribution of customers by gender.

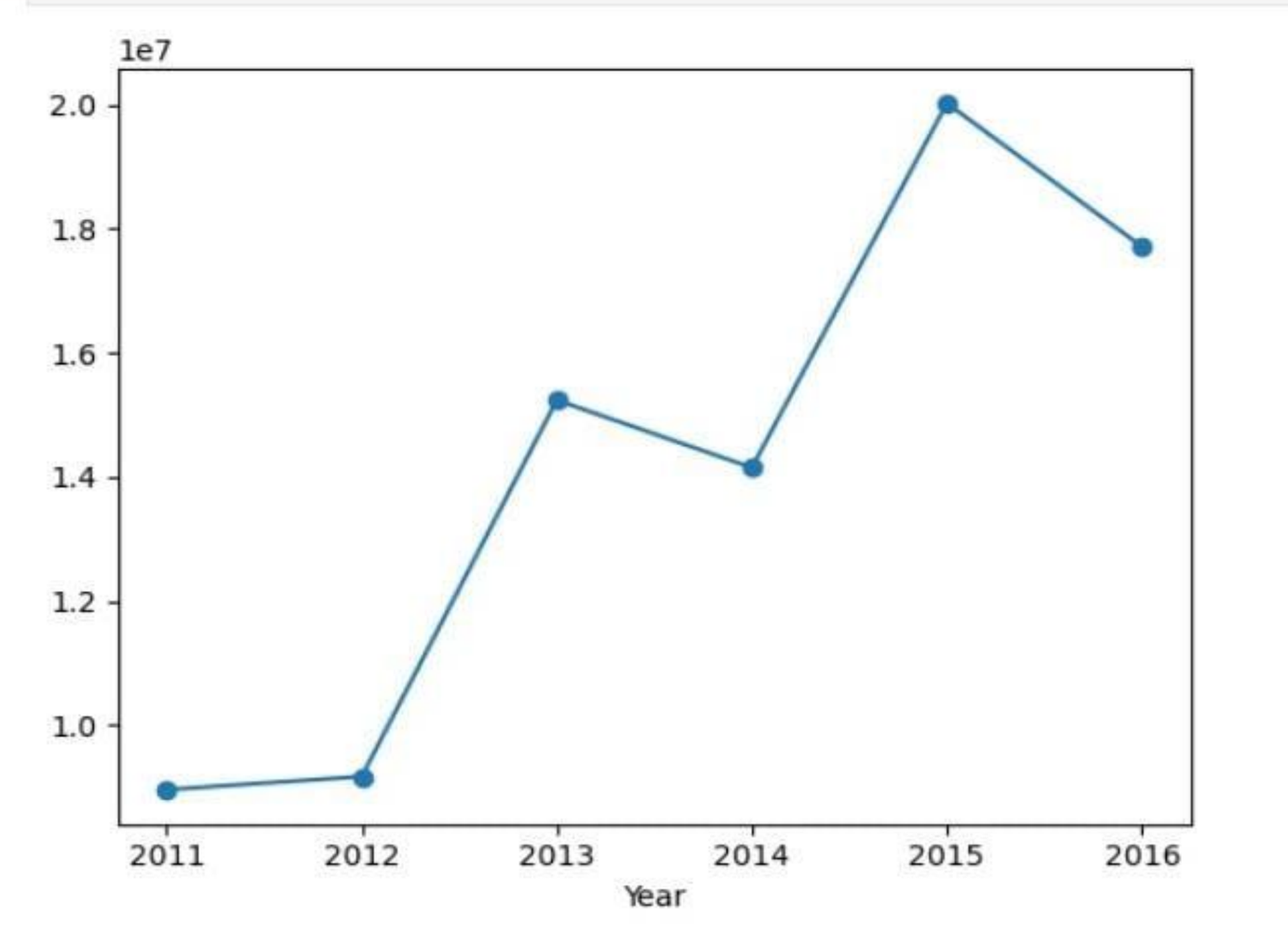
Revenue by Age Group:
Total revenue generated by each age group.

Data Visualization

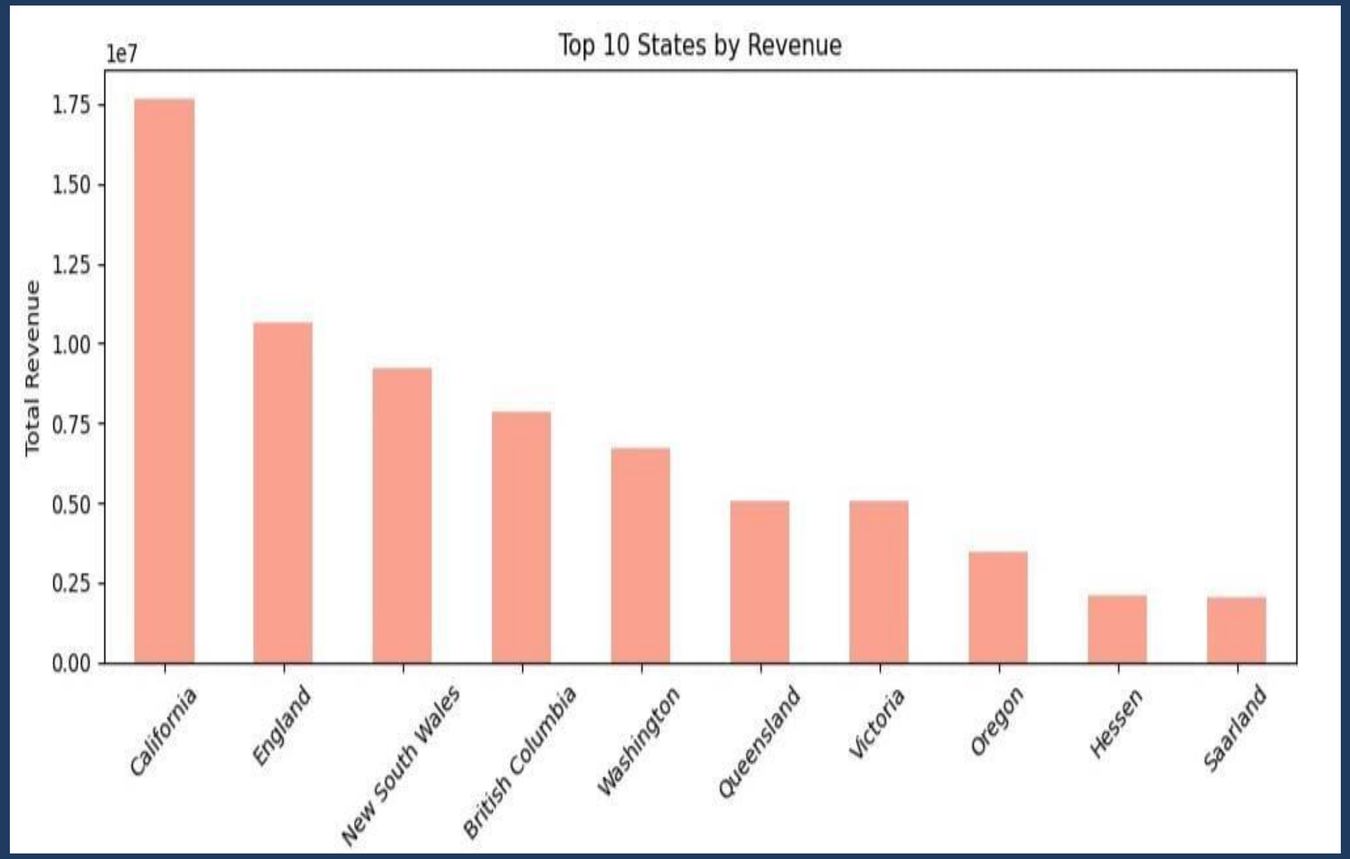
The United States generates the highest revenue among all countries, followed by Australia and the United Kingdom, while Canada records the lowest revenue in this dataset.



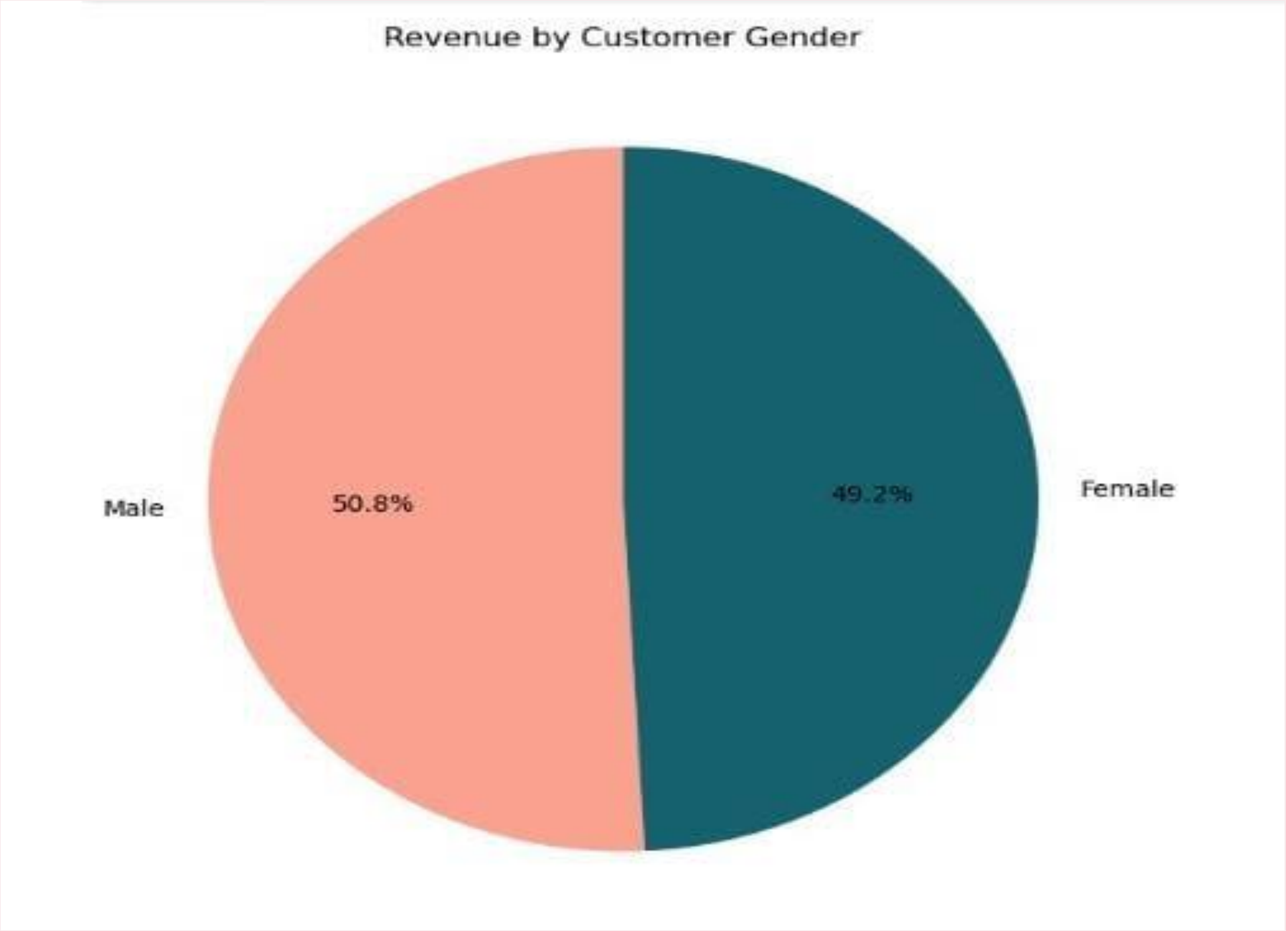
Revenue peaks in 2015, followed by 2016, while earlier years (2011–2012) show the lowest performance, indicating a general growth trend over time with the highest sales achieved in the later years.



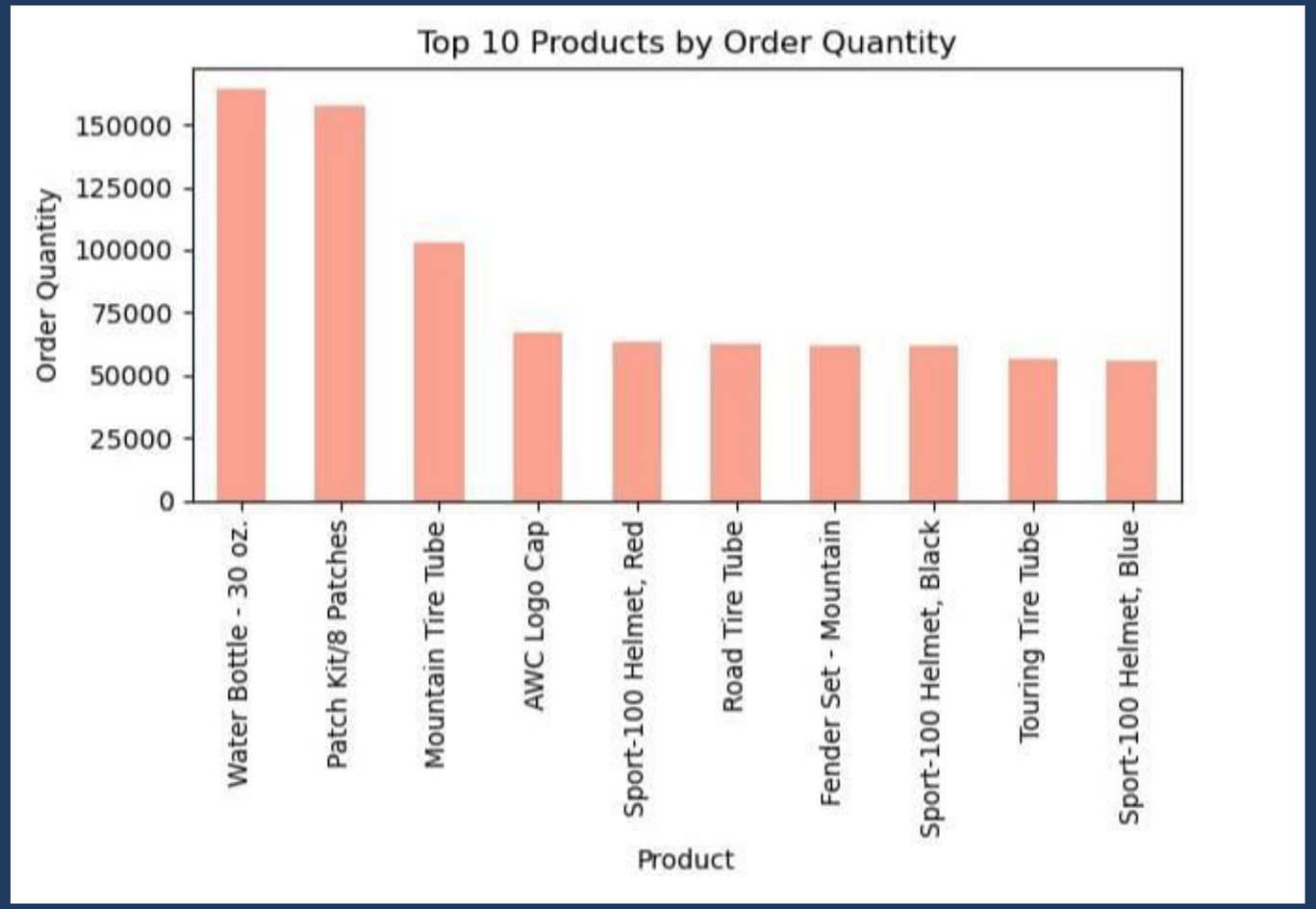
California generates the highest revenue among all states, followed by England and New South Wales, indicating these regions are the strongest markets in terms of sales performance.



Male customers generate slightly higher revenue than female customers, indicating a relatively balanced contribution from both genders with a small difference in favor of males.

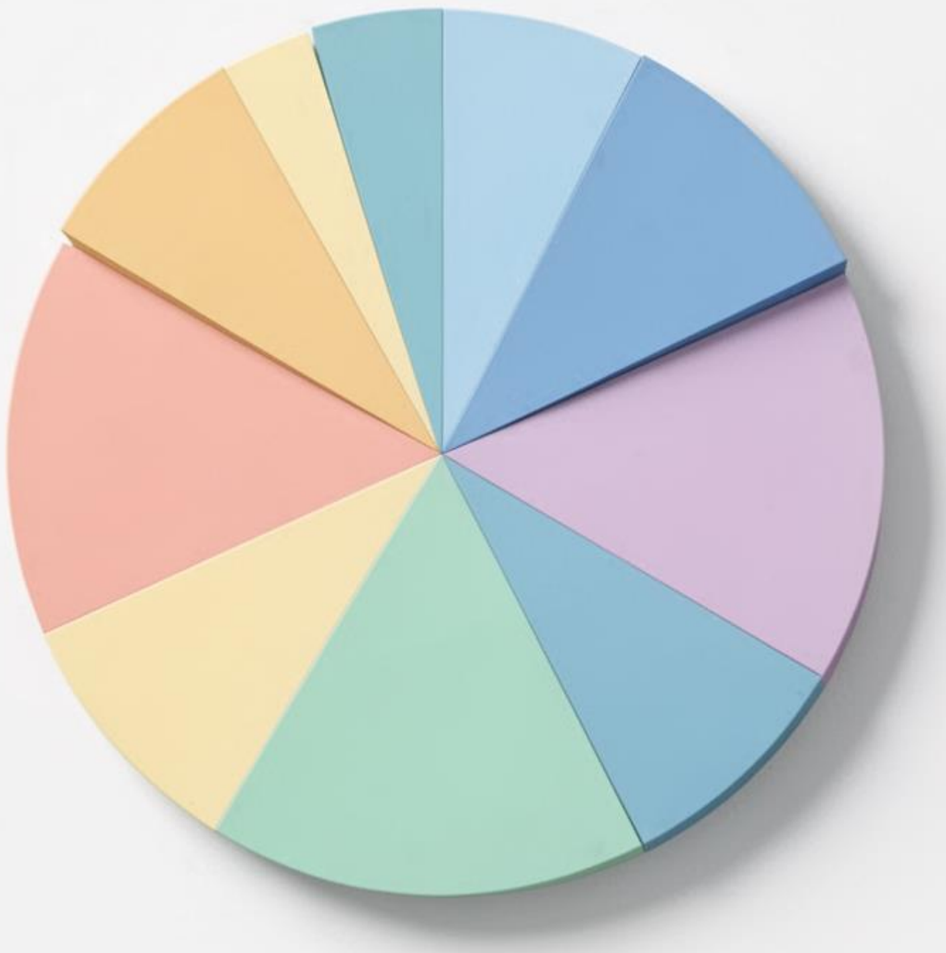


Water Bottle - 30 oz. and Patch Kit/8 Patches are the most frequently ordered products, followed by tire tubes and helmets, indicating that low-cost accessories and essential maintenance items drive the highest order quantities compared to other products.



Segment Analysis

Revenue analysis by country and state
Revenue comparison across age groups
Revenue distribution by product category and sub-category
Customer segmentation based on gender
Revenue trends across year and month
Order quantity distribution across product categories



Recommendations for Growth

Focus on high-revenue countries to maximize sales performance.

Target age groups with highest revenue contribution.

Promote top-performing product categories.

Optimize marketing strategies based on customer segments (gender & location).

Increase focus on high-demand periods (year/month trends).

Limitations of Analysis

The dataset includes only internal variables (Revenue, Profit, Quantity).

No data on external factors affecting sales (e.g., market conditions) .

Analysis depends on available time fields (Year, Month, Day) without deeper historical context.